Dyad Geopolitical Scores – Variable Codebook

Dataset Overview

File: `dyad_geopolitical_scores_Major24xAll.csv`

Unit of observation: Dyad-year (undirected, alphabetically ordered by ISO 3166-1 alpha-3 code)

Time coverage: 1950–2024 (75 years)

Country coverage: 4,332 dyads in which at least one country belongs to the 24 major nations. Each of the 24 major nations is paired with all 193 UN member states.

Total observations: 324,900 (4,332 dyads × 75 years)

Panel type: Balanced (all dyad-year cells present; zero-event cells recorded as 0)

Source data: Global Geopolitical Events (GGE) Database, v1

Scope note: This file covers the sample used in *Fan (2025), "Measuring Geopolitical Alignment and Economic Growth"* – all bilateral pairs involving at least one of the 24 major nations. The full-coverage panel (all 18,528 UN-member dyads) will be released alongside the companion measurement paper, *Fan, Liu, and Xiang (2026), "The Structure of International Order, 1950–2024."*

Major 24 Nations

Code	Country	Code	Country	Code	Country
ARG	Argentina	ESP	Spain	KOR	South Korea
AUS	Australia	FRA	France	MEX	Mexico
BEL	Belgium	GBR	United Kingdom	NLD	Netherlands
BRA	Brazil	IDN	Indonesia	POL	Poland
CAN	Canada	IND	India	RUS	Russia
CHE	Switzerland	ITA	Italy	SAU	Saudi Arabia
CHN	China	JPN	Japan	TUR	Turkey
DEU	Germany	DNK	Denmark	USA	United States

Variable Definitions

Identifiers

Variable	Type	Description
`dyad`	String	Country pair identifier in alphabetical order (e.g., `CHN-USA`, `DEU-FRA`)
`country1_code`	String	ISO3 code of the first country (alphabetically earlier)
`country2_code`	String	ISO3 code of the second country
`year`	Integer	Calendar year (1950–2024)

Primary Geopolitical Measures

Variable	Type	Range	Description
`geo_score`	Float	$[-1, 1]$	**Static geopolitical score.** Annual average of Goldstein Scale values divided by 10. Missing if no events observed.
`geo_score_dyn`	Float	$[-1, 1]$	**Dynamic geopolitical score.** Exponentially weighted score incorporating historical information with depreciation rate $\delta = 0.3$. Never missing; decays toward 0 during periods without events.
`geo_score_ma`	Float	$[-1, 1]$	**Moving average geopolitical score.** Event-weighted average of `geo_score` over a 3-year window ($t - 2$ to t). Missing if no valid observations in window.

Auxiliary Variables

Variable	Type	Range	Description
`number_of_events`	Integer	$[0, \infty)$	Count of bilateral geopolitical events in the given year
`cum_events`	Float	$[0, \infty)$	Effective cumulative event count with exponential depreciation
`weight`	Float	$[0, 1]$	Updating weight $\phi_{ij,t}$ assigned to the current year's score in the dynamic calculation

Score Construction

Static Score

Let $\{s^n_{ij,t}\}_{n=1}^{\tilde{N}_{ij,t}}$ denote the Goldstein scores for the set of events between countries i and j in year t, where $\tilde{N}_{ij,t}$ is the number of events. The yearly average event score, normalized to the $[-1, +1]$ interval, is:

$$\tilde{S}_{ij,t} = \frac{1}{\tilde{N}_{ij,t}} \sum_{n=1}^{\tilde{N}_{ij,t}} s^n_{ij,t} \,/\, 10$$

This corresponds to the variable `geo_score`. It is missing (NaN) for dyad-years with no observed events.

Dynamic Score

The dynamic score balances the contemporaneous impact of political events with the institutional memory that characterizes interstate relations. When events are observed ($\tilde{N}_{ij,t} > 0$):

$$S_{ij,t} = (1 - \phi_{ij,t}) \cdot S_{ij,t-1} + \phi_{ij,t} \cdot \tilde{S}_{ij,t}$$

where the updating weight and effective cumulative event count are:

$$\phi_{ij,t} = \frac{\tilde{N}_{ij,t}}{N_{ij,t}}, \qquad N_{ij,t} = (1 - \delta)\, N_{ij,t-1} + \tilde{N}_{ij,t}$$

When no events are observed ($\tilde{N}_{ij,t} = 0$), the update equation does not apply; instead, both the score and effective event count decay:

$$S_{ij,t} = (1 - \delta)\, S_{ij,t-1}, \qquad N_{ij,t} = (1 - \delta)\, N_{ij,t-1}$$

Initialization. For the first year of each dyad ($t = t_0$): $S_{ij,t_0} = \tilde{S}_{ij,t_0}$ and $N_{ij,t_0} = \tilde{N}_{ij,t_0}$. When the initial year has no events, $S_{ij,t_0} = 0$ and $N_{ij,t_0} = 0$.

The depreciation rate is set to $\delta = 0.3$. The updating weight $\phi_{ij,t}$ determines how responsive the score is to new information: country pairs with long interaction histories have large $N_{ij,t}$, so any single year's events receive a small weight, reflecting the difficulty of fundamentally shifting a deeply established relationship. Conversely, pairs with sparse histories update more rapidly.

This corresponds to the variable `geo_score_dyn`. It is never missing; it equals 0 for dyads with no prior events and decays toward 0 during periods without events.

Moving Average Score

The moving average score is an event-weighted average of `geo_score` over a 3-year window ($t - 2$ to t):

$$S^{\text{ma}}_{ij,t} = \frac{\sum_{k=t-2}^{t} \tilde{N}_{ij,k} \cdot \tilde{S}_{ij,k}}{\sum_{k=t-2}^{t} \tilde{N}_{ij,k}}$$

where the sum includes only years with non-missing scores. Missing if no events occur in the entire window.

Key Parameters

Parameter	Value	Description
δ	0.3	Depreciation rate for dynamic scores and cumulative events
MA window	3 years	Moving average window ($t-2$ to t)

Goldstein Scale Interpretation

The underlying Goldstein Scale ranges from -10 (maximum conflict) to $+10$ (maximum cooperation). All `geo_score` variants are normalized by dividing by 10:

Score range	Interpretation
-1.0 to -0.6	Severe conflict (military attacks, war, use of force)
-0.6 to -0.3	Significant hostility (military threats, sanctions, breaks in relations)
-0.3 to 0.0	Minor conflict (diplomatic protests, recalls, verbal hostility)
0.0 to $+0.3$	Neutral to mild cooperation (routine diplomatic exchanges, visits)
$+0.3$ to $+0.6$	Moderate cooperation (agreements, treaties, joint statements)
$+0.6$ to $+1.0$	Strong cooperation (military alliance, economic integration, extensive aid)

Event Data Methodology

Overview

The underlying event data were compiled by a large language model (Gemini 2.5 Pro) following a structured prompt. Each event represents a distinct significant bilateral political interaction between two countries in a given year.

Total events in the GGE database: 833,485 across all 18,528 dyads among 193 UN member states, 1950–2024.

Event Identification Pipeline

The LLM follows a 5-step analytical procedure for each dyad-year:

1. **Entity verification.** Confirm that both countries existed as the specified political entities in the target year; if not, identify the appropriate historical predecessor (e.g., Soviet Union before December 1991).

2. **Systematic search.** Search across all six event domains (below) using official government sources, international organizations, major news archives, and academic sources.
3. **Event coding.** Classify each event by category (A1–F7), assign CAMEO quad class and root code, and calibrate a Goldstein score (−10 to +10).
4. **Relationship assessment.** Assign an overall annual assessment using a 9-point ordinal scale ranging from *State of War / Active Conflict* to *Alliance*.
5. **Structured output.** Generate a validated JSON object with 17 required fields per event.

Event Taxonomy

Events are classified into 6 top-level domains with 23 subcategories:

A. Economic Relations (6 subcategories)

- A1. Trade Policy and Market Access – tariffs, non-tariff barriers, trade agreements
- A2. Financial and Monetary Relations – financial sanctions, monetary cooperation, investment controls
- A3. Economic Coercion and Incentives – comprehensive sanctions, export controls, economic aid
- A4. Strategic Economic Sectors – energy, technology/digital, critical resources
- A5. Economic Integration and Infrastructure – BRI/competing programs, supply chains, regional arrangements
- A6. Other Economic Events – market access issues, boycotts, gray zone actions

B. Diplomatic and Political Relations (4 subcategories)

- B1. Formal Diplomatic Engagement – embassy actions, ambassador recalls, diplomatic protests
- B2. High-Level Political Interactions – summits, ministerial meetings, strategic dialogues
- B3. Public Diplomacy and Rhetoric – policy speeches, parliamentary resolutions, propaganda
- B4. Cultural and Educational Diplomacy – cultural agreements, scholarship programs, academic exchanges

C. Security and Defense (3 subcategories)

- C1. Military Cooperation and Competition – defense agreements, arms sales, joint exercises
- C2. Security Incidents and Responses – border skirmishes, airspace violations, territorial claims
- C3. Intelligence and Cyber – espionage, cyber attacks, intelligence sharing

D. Legal, Territorial, and Movement (3 subcategories)

- D1. International Legal Actions – ICJ/WTO cases, extradition, legal cooperation
- D2. Territorial and Maritime Issues – EEZ disputes, border demarcation, sovereignty claims
- D3. Movement of People – visa policies, migration agreements, travel bans

E. Multilateral and Global Governance (2 subcategories)

- E1. International Organizations – UN system, regional organization actions
- E2. Global Issues Cooperation/Competition – climate, health, human rights

F. Other Significant Events (5 subcategories)

- F1. Historical and Symbolic Events – apologies, memorial visits, symbolic gestures
- F2. Natural Disasters and Humanitarian Crises – disaster response, crisis management

- F3. Sports and Major Events – Olympic boycotts, major event hosting
- F4. Technology and Space – joint missions, technology disputes
- F5–F7. Environmental, Communications, and Other – transboundary issues, media relations, other interactions

CAMEO Framework

Each event is classified using the Conflict and Mediation Event Observations (CAMEO) framework into one of four quad classes:

Quad class	CAMEO root codes	Examples
Verbal Cooperation	01–05	Public statements, appeals, diplomatic cooperation
Material Cooperation	06–08	Material aid, economic cooperation, concessions
Verbal Conflict	09–14	Demands, disapproval, threats, protests
Material Conflict	15–20	Force posture, reduced relations, coercion, assault

The 20 CAMEO root codes provide a standardized two-digit classification of event types, enabling comparison with other political event datasets.

Goldstein Scale

The Goldstein scale assigns each event a cooperation–conflict intensity score:

- **−10 to −8:** Highly conflictual (military attacks, mass violence)
- **−8 to −5:** Threats, coercion, severe sanctions
- **−5 to −2:** Diplomatic protests, rejections, minor sanctions
- **−2 to 0:** Mild disagreements, negative rhetoric
- **0 to +2:** Initial positive gestures, routine exchanges
- **+2 to +5:** Consultations, cooperative intent, diplomatic meetings
- **+5 to +8:** Significant aid, major agreements, material cooperation
- **+8 to +10:** Major concessions, peace agreements, alliance formations

Scores are calibrated against CAMEO root codes and adjusted for bilateral context.

Usage Notes

1. **Balanced panel.** The dataset includes all 4,332 dyad-year combinations within the time range, including years with no events (`number_of_events = 0`).

2. **Dyad directionality.** Dyads are undirected. The two ISO3 codes are always arranged alphabetically (e.g., `CHN-USA`, never `USA-CHN`).

3. **Historical entities.** Country codes reflect present-day ISO3 assignments. Events involving predecessor states (e.g., the Soviet Union before 1991) are coded under the successor state's code (e.g., `RUS`).

4. **Recommended usage:**

   - Use `geo_score_dyn` for cross-sectional analysis, as it reflects the cumulative geopolitical alignment status of each dyad at a given point in time.
   - Use `geo_score_dyn` for time-series analysis, with `geo_score` as a robustness check to verify that results are not driven by the smoothing procedure.
   - Use `geo_score_ma` as an additional robustness check or to reduce measurement error from year-to-year volatility.

5. **Interpretation of sign:**

   - Negative values indicate conflict, hostility, or deteriorating relations.
   - Positive values indicate cooperation, partnership, or improving relations.
   - Values near zero indicate neutral or minimal interaction.

References

Tianyu Fan (2025). "Measuring Geopolitical Alignment and Economic Growth." *Working Paper, Yale University.* arXiv:2507.04833

Tianyu Fan, Mai Wo, and Wei Xiang (2025). "Geopolitical Barriers to Globalization." *Working Paper, Yale University.* arXiv:2509.12084

Tianyu Fan, Jizhou Liu, and Wei Xiang (2026). "The Structure of International Order, 1950–2024." *Working Paper, Yale University. Coming soon.*